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COMP4139
Machine Learning (MLE)

- Unsupervised Learning (Cluster Analysis)

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Recap

- How do we evaluate ML models?
- What is overfitting and underfitting?
- What is precision? TP/TP+FP
- What is recall? TP/TP+FN
- What is recall also called?
- What is specificity? TN/TN+FP
- How do we know if the results we have got are statistically significant?

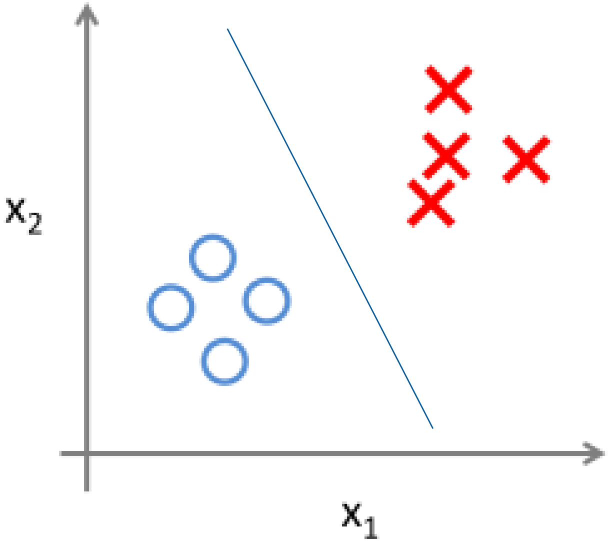
Recap

		Real (Actual, Observed)		
		Real Negatives **TN+FP**	**Real Positives** **TP+FN**	
Predicted	**Predicted Negatives** **TN+FN**	true negatives (TN)	false negatives (FN)	
	Predicted Positives **TP+FP**	false positives (FP)	true positives (TP)	**Precision** = true positives/**PRE**di**C**ted positives TP/(TP+FP)
		Specificity **SPIN** (**SP**ecificity **I**s **N**egative) true negatives/real negatives TN/(TN+FP)	**Sensitivity** **SNIP** (**S**e**N**sitivity **I**s **P**ositive) true positives/real positives TP/(TP+FN)	**Accuracy** **true** predictions/**all** predictions (TP+TN)/(TP+TN+FP+FN)
			Recall true positives/**REAL** positives TP/(TP+FN) Recall = Sensitivity	

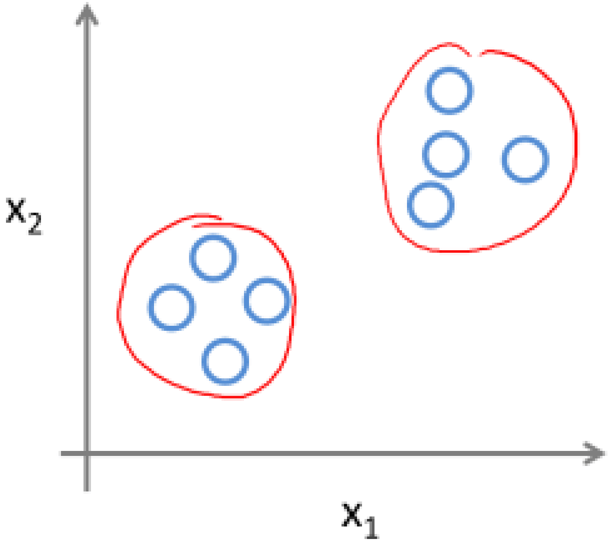
Learning Objectives

- What is unsupervised learning?
- Similarity between vectors
- K-Means Clustering
- Agglomerative Hierarchical Clustering
- Density-based Spatial Clustering of Applications with Noise (DBSCAN)
- Expectation Maximisation

Unsupervised Learning



Supervised Learning



Unsupervised Learning

Unsupervised Learning

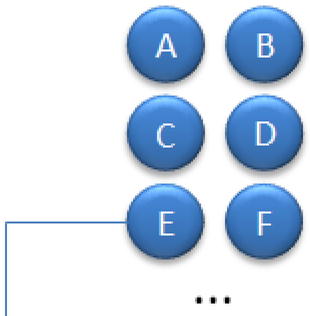
- A type of algorithm that learns hidden patterns from unlabelled data.

- Methods:
  - **Cluster Analysis**: divide data into meaningful groups
  - **Dimensionality reduction**: Principal Component Analysis, Autoencoder, Generative models, etc.
  - Most cases we combine the above two.

- Applications:
  - Google: 60 trillion webpages are clustered into sub-categories.
  - Biology: groups of genes that have similar functions.
  - Business: personalised products marketing.

Clustering Elements

Objects



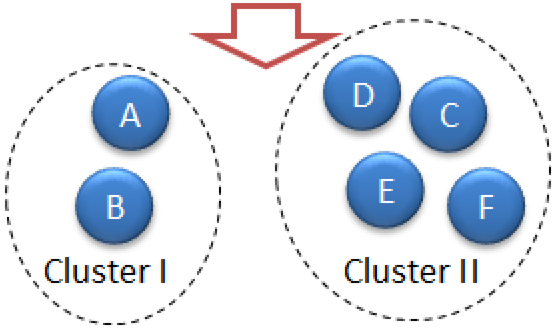
Features*:
$x_1, x_2, x_3, x_4, \ldots$

1. Feature selection/ construction
- all features*
- PCA, MDS, ICA, tSNE
- a priory knowledge

(*) Additionally, features could be transformed to ensure reasonable importance of each feature, e.g:
- log-transformation,
- standardization,
- ranking, etc.

2. Distance
- Euclidean
- correlation: 1 - *r*
- Manhattan
- cosine (inner product)

3. Clustering
- hierarchical
- k-means
- density-based (DBSCAN)



Key Elements of Cluster Analysis

- Features that describe the subjects.
  - Normally high dimensional feature vectors.
  - We can apply PCA, manifold learning or deep learning based methods to firstly reduce the dimensionality of the feature.

- Similarity functions
  - Euclidean, Cosine, Manhattan and Jaccard etc.

- Basic clustering algorithms
  - Centroid model (e.g. K-means)
  - Connectivity model (e.g. Agglomerative hierarchical)
  - Density model (e.g. DBSCAN)
  - Distribution model (e.g EM)

- Cluster validation

Similarity Function

- Given two column vectors **A** and **B** that contains features of two data samples.

- Euclidean distance
  - Squared root of the summed difference of A and B.

$$ED = \sqrt{(A-B)^T (A-B)}$$

- Cosine distance
  - Larger value indicates A and B share similar direction

$$CD = \cos(\theta) = \frac{A \cdot B}{\|A\| \, \|B\|}$$

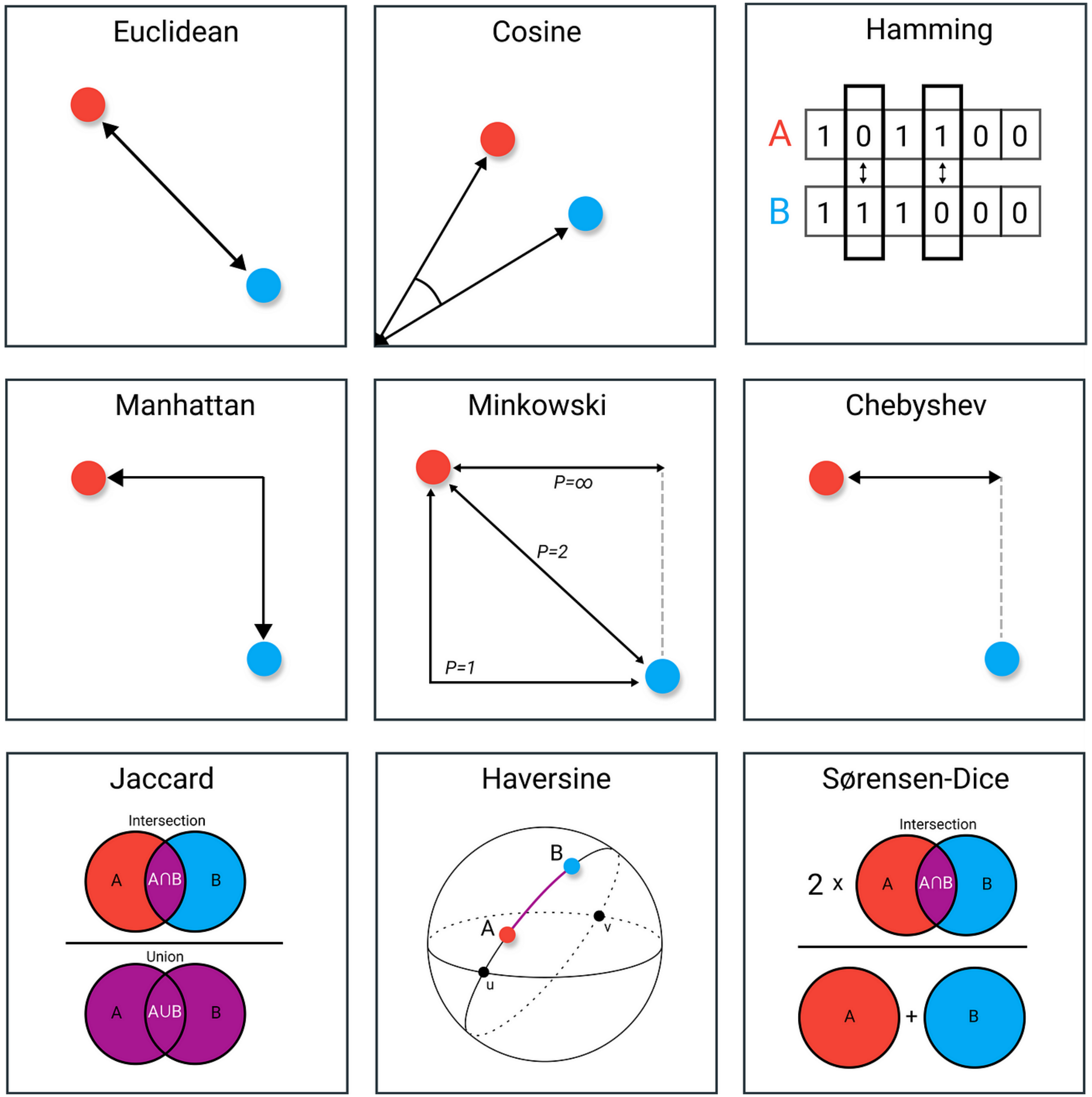
- Manhattan distance
  - Mean absolute difference of A and B.

$$MD = \frac{1}{N} \sum^{N} |A - B|$$

- Jaccard distance
  - Overlapping over union of sets A and B.

$$JD = \frac{|A \cap B|}{|A \cup B|}$$

Similarity Function



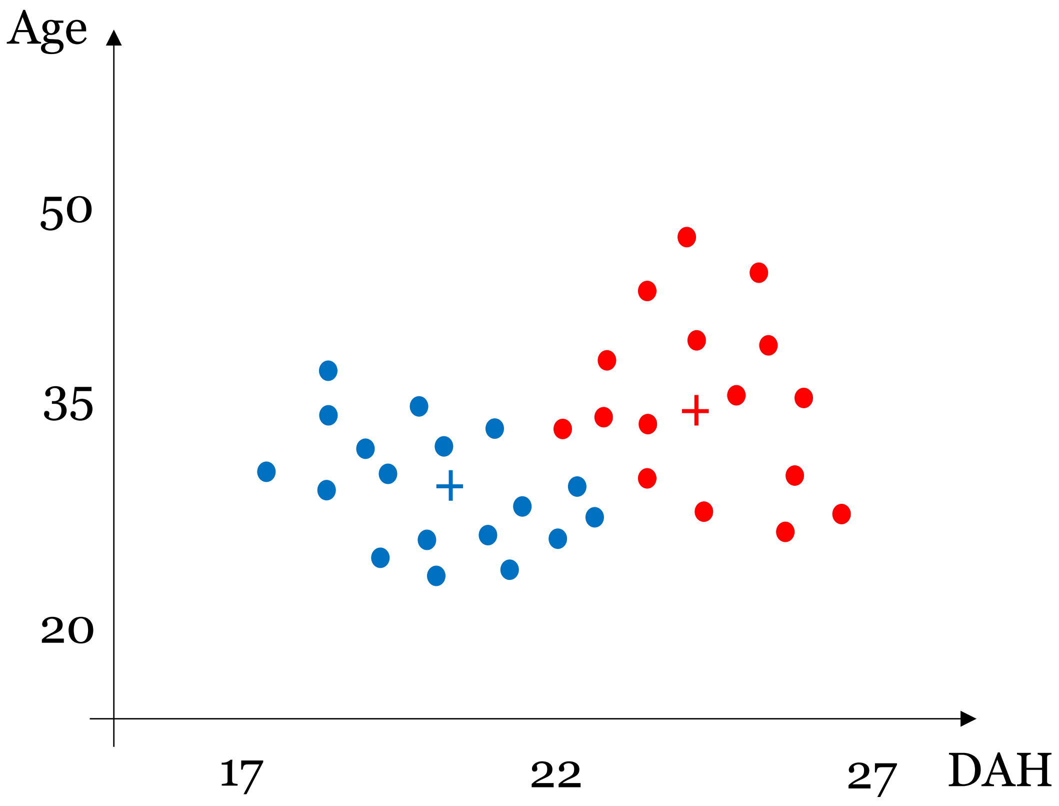
- Euclidean - Continuous, dense data where absolute magnitude matters. Ex. Distance between houses
- Manhattan - Data with a grid or lattice structure, or when dimensions are not equally important. Ex. Calculating the distance in a city where you must travel along streets
- Cosine - High-dimensional and sparse data, especially text, where direction (orientation) is more important than magnitude. Ex. Document similarity
- Jaccard - Comparing the similarity between two sets, particularly with binary or categorical data. Ex. Comparing the set of websites visited by two users to see how many sites they have in common.

Example Data

K-means Algorithm

Algorithm:

- Randomly select K points as the initial centroids for each of the K groups.
- *Repeat*
  - Assigning each point to its closest centroid.
  - Re-compute the centroid of each cluster.
- *Until* centroids do not change.

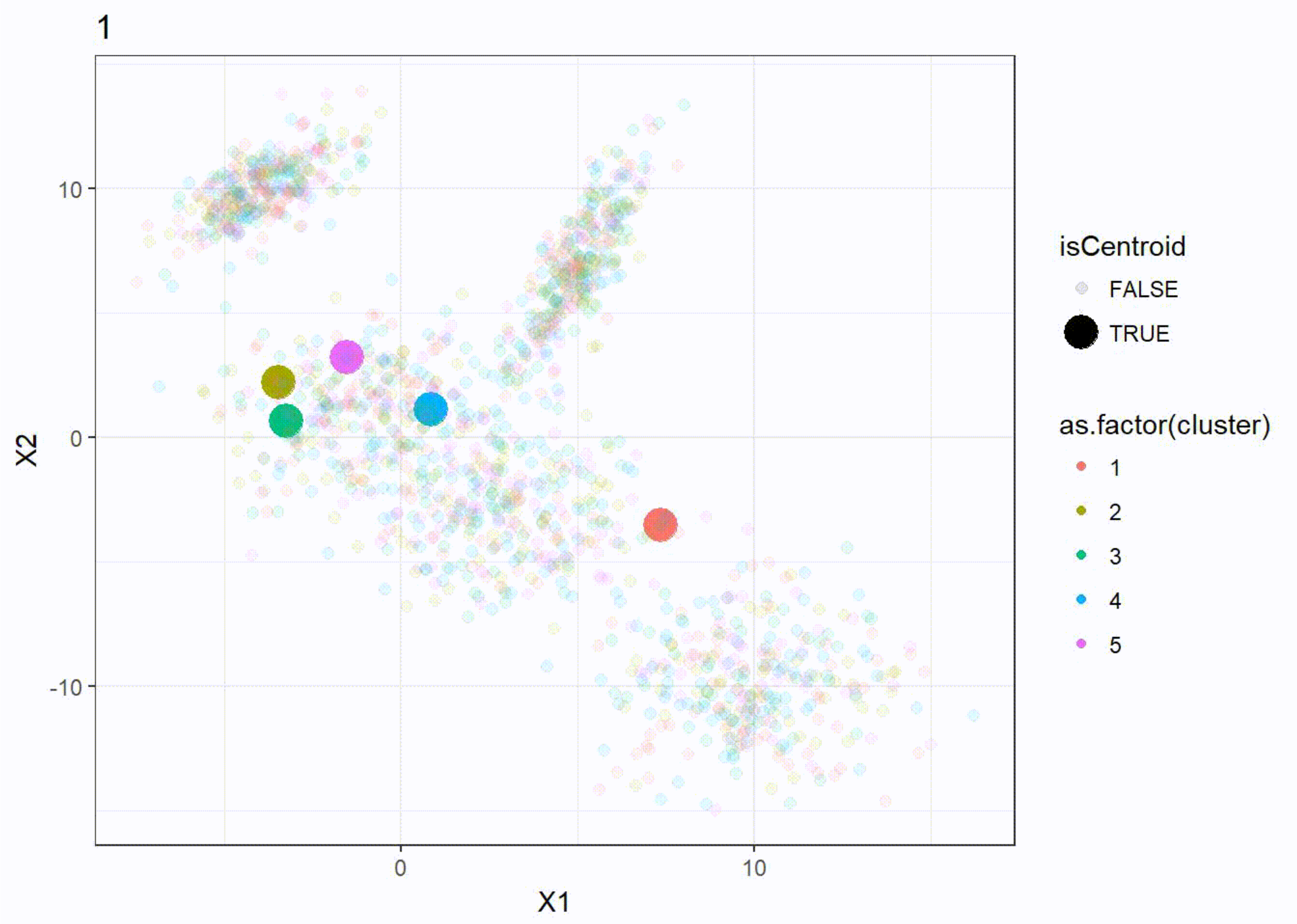


K-means

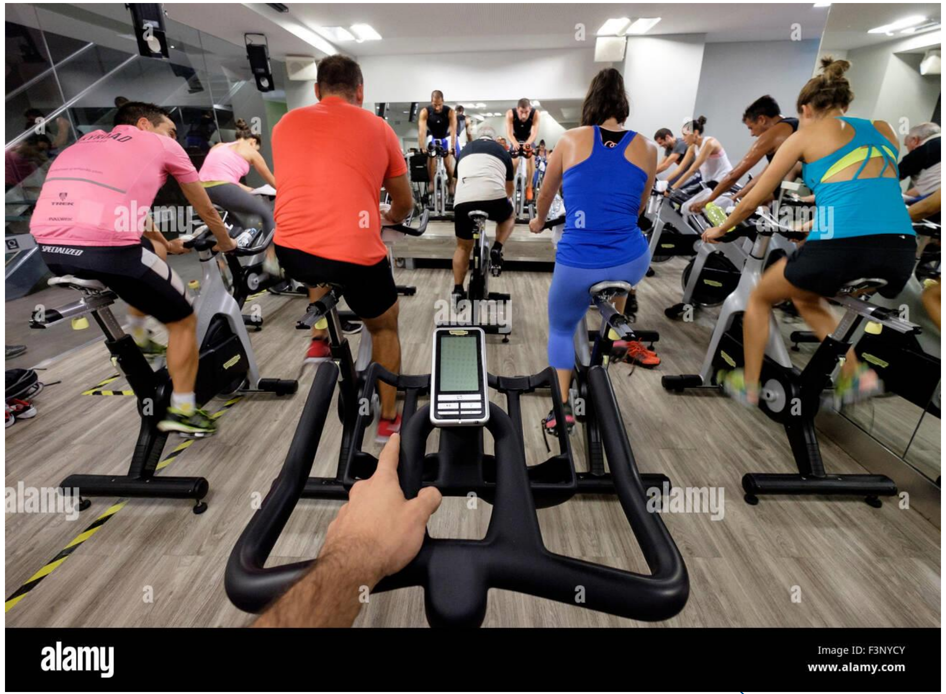
Pros/Cons

✓ Simple and efficient
✓ Scalability
✓ Fast convergence
× Solution dependent on the initialisation
× Need to specify number of clusters
× Sensitive to outliers
× Curse of dimensionality

K-Means Visualization



Outcome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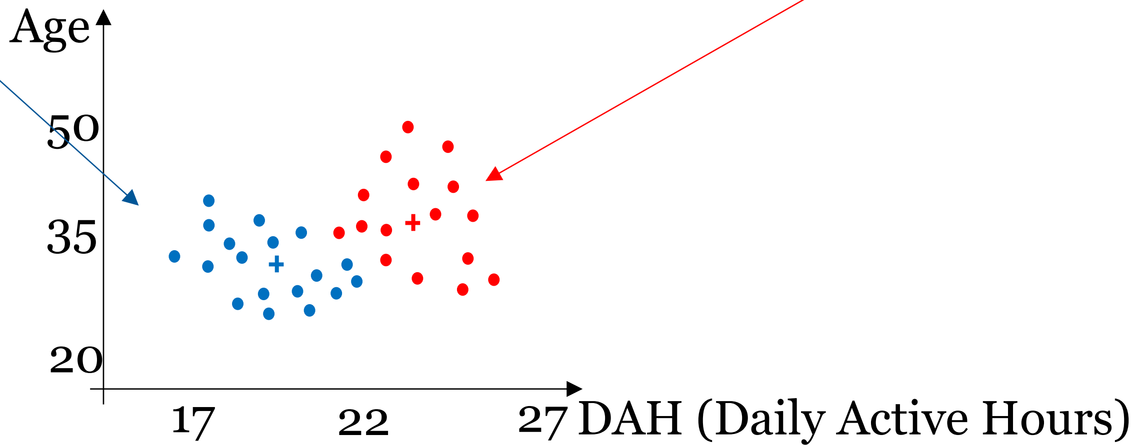
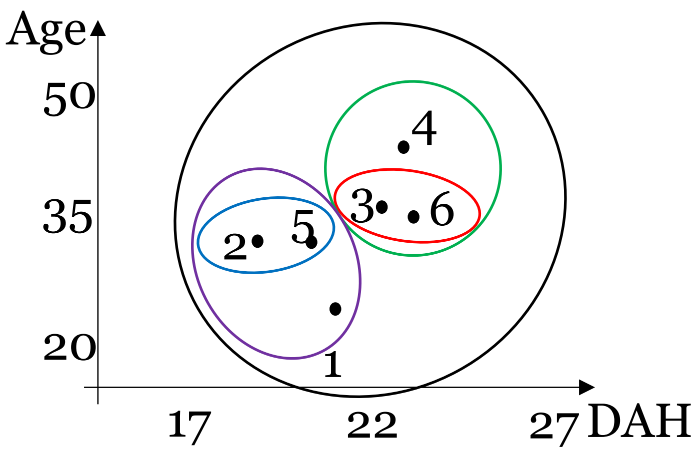
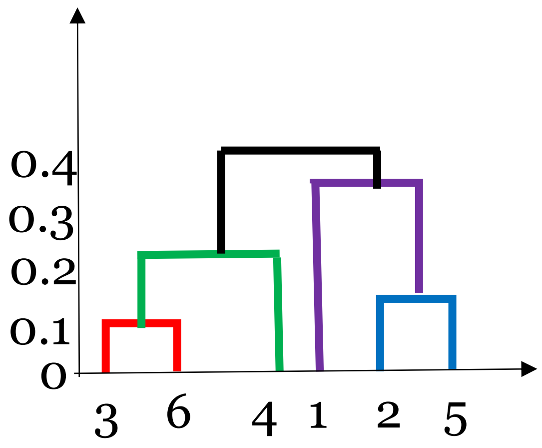

Age
50
35
20
17
22
27
DAH (Daily Active Hours)

Agglomerative Hierarchical Clustering

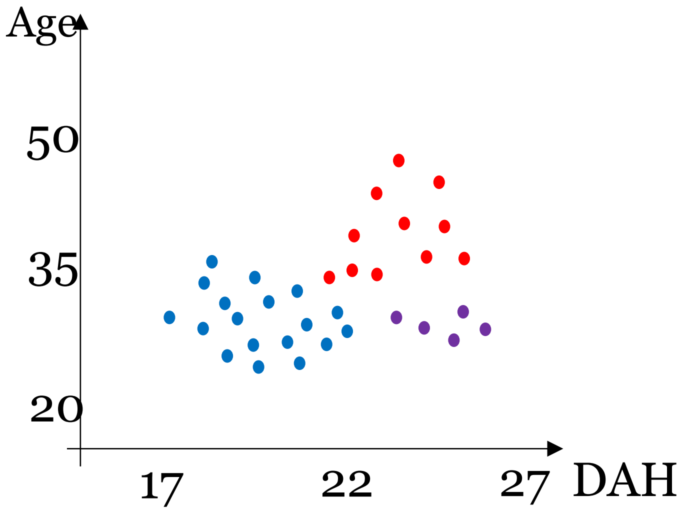
Algorithm:

- Treat each data as a cluster. Compute the similarity matrix between each pair of data.
- *Repeat*
  - Merge the closest two clusters.
  - Update the similarity matrix.
- *Until* only one cluster remains.





	P1	P2	P3	P4	P5	P6
P1	0	0.24	0.22	0.37	0.34	0.23
P2		0	0.15	0.20	0.14	0.25
P3			0	0.15	0.28	0.11
P4				0	0.29	0.22
P5					0	0.39
p6						0

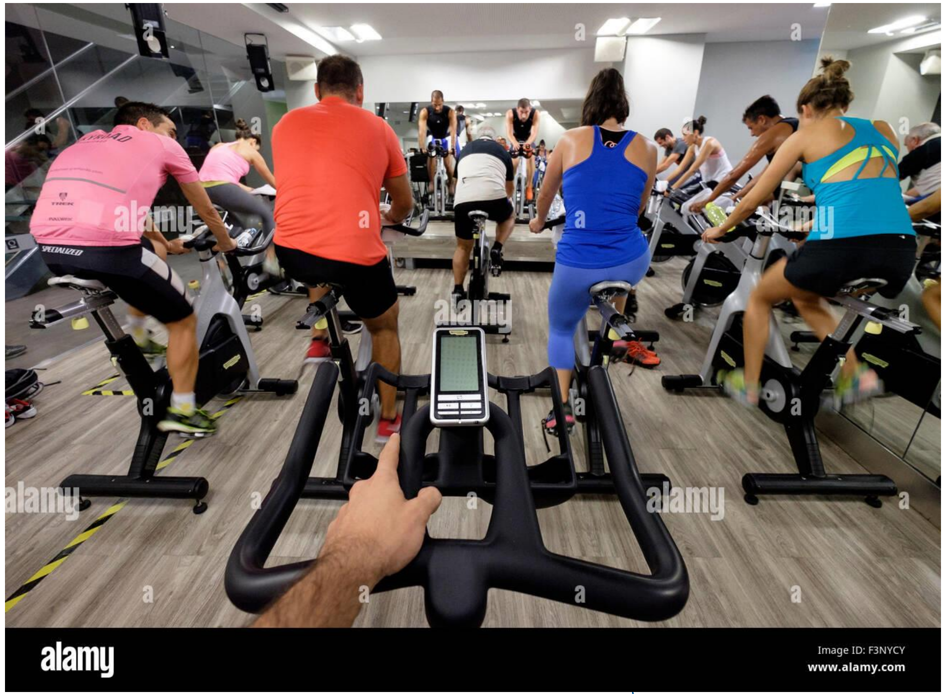


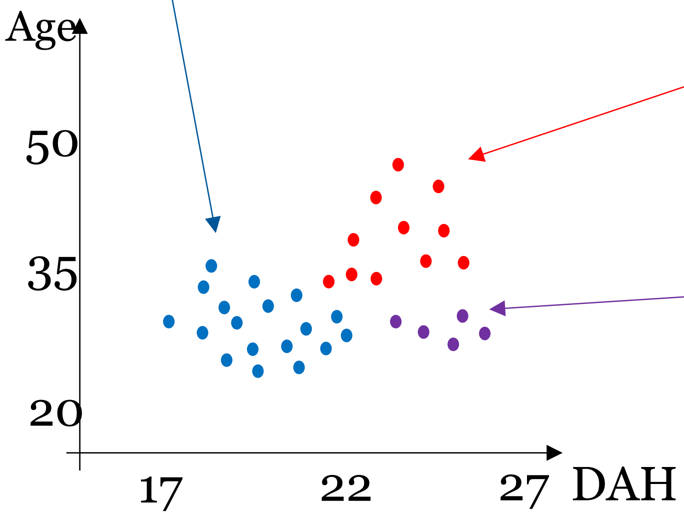
Hierarchical Clustering

Pros/Cons

✓Flexible with number of clusters, no need for pre-determined ‘k’
✓Can capture hierarchical relationship
✓Interpretability
✓Handles non-elliptical shapes
× Solution is local optimum, dependent on subject functions (e.g. min, max, group average etc.)
× High computational cost in terms of speed
× Requires larger memory. Ex. 10,000 data points need 10000x10000 matrix stored somewhere

Outcome





Example Data

Density-based Spatial Clustering of Applications with Noise (DBSCAN)

Key concepts:

1.Core Points

A point is called a core point if it has enough neighboring points within a certain distance.

Rule: If a point has at least MinPts points (including itself) within a radius ε (epsilon), it's a core point.

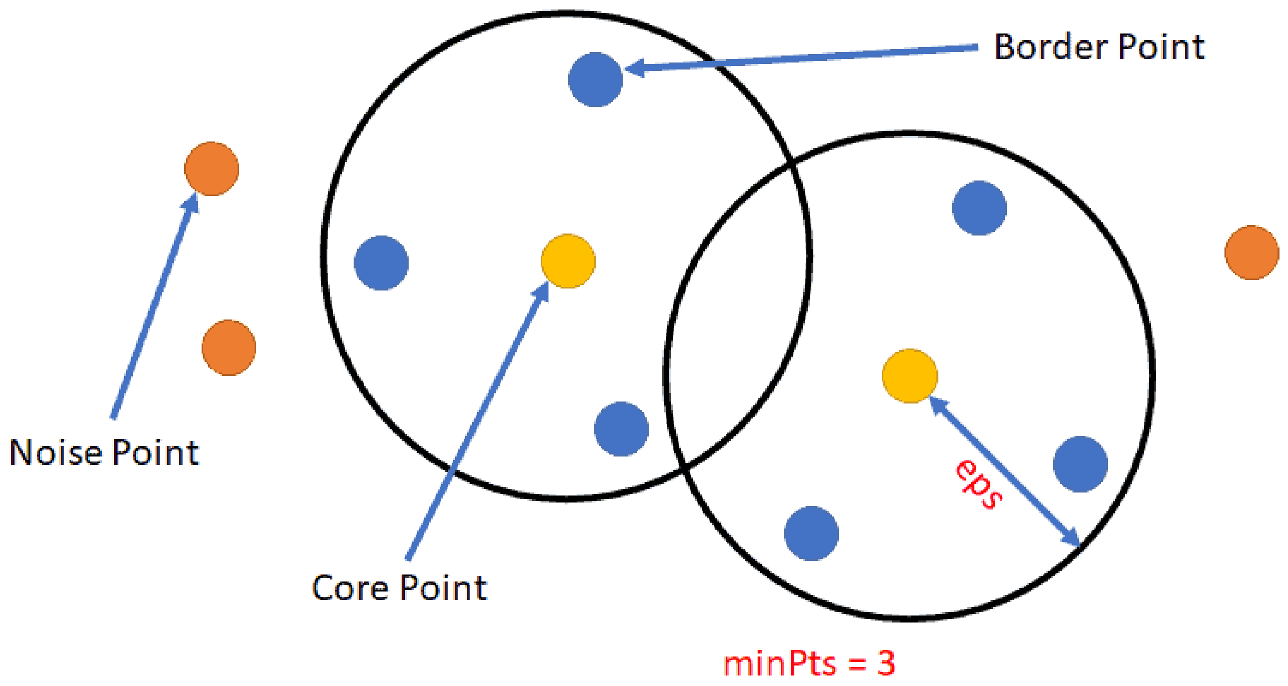
2. Border Points

A border point is close to a core point but doesn't have enough neighbors to be a core point itself.

Rule: It has fewer than MinPts within ε, but is within ε of a core point.

3. Noise Points

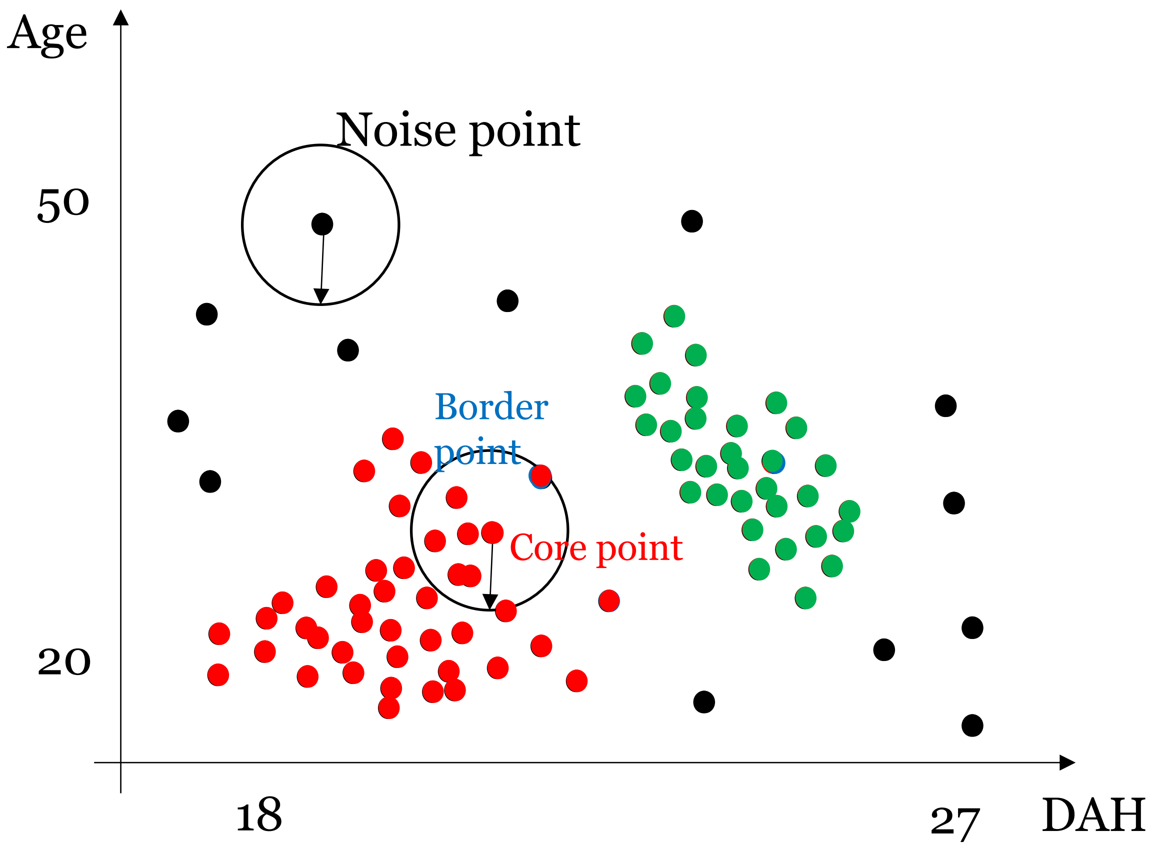
A noise point is isolated, it's neither a core point nor close enough to any core point. These are treated as outliers.



Density-based Spatial Clustering of Applications with Noise (DBSCAN)

Algorithm:

- Find the neighbourhood points of every point with distance E.
- Identify the core points with more than minPoints neighbours
- Connect the core points if they are within distance E.
- Make each group of connected core points into a separate cluster.
- Assign each non-core point to a nearby cluster if they are within distance E, called border points.
- Unassigned points are noise points.



Density-based Spatial Clustering of Applications with Noise (DBSCAN)

Pros/Cons

✓ Robust to outliers
✓ Learn non-regular population density patterns
✓ Arbitrary cluster shapes
✓ Automatically determine the number of clusters
× Non robust to variable density clusters (due to single E is used)
× Computationally expensive
× Sensitive to parameter settings

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Half-term Course Feedback

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Labs (start from w/c 06th October 2024):

- **Tuesday 11 am - 1 pm** Lab sessions in CS-A32

MS-Teams Online Support (Join code: ddak23n):

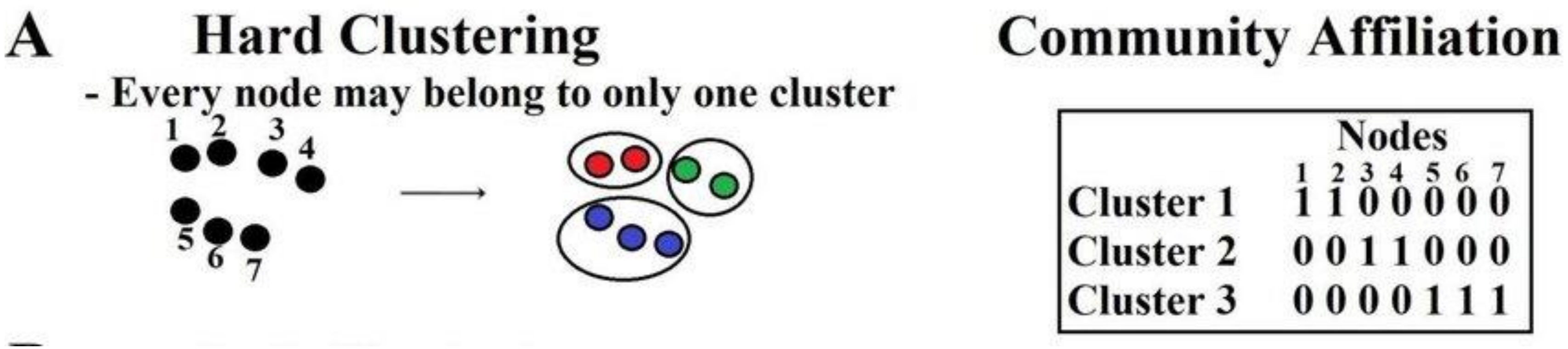
- **It is used for organising groups for coursework**
- **It is used for Q&As. Not for lecture delivering.**

Assessment

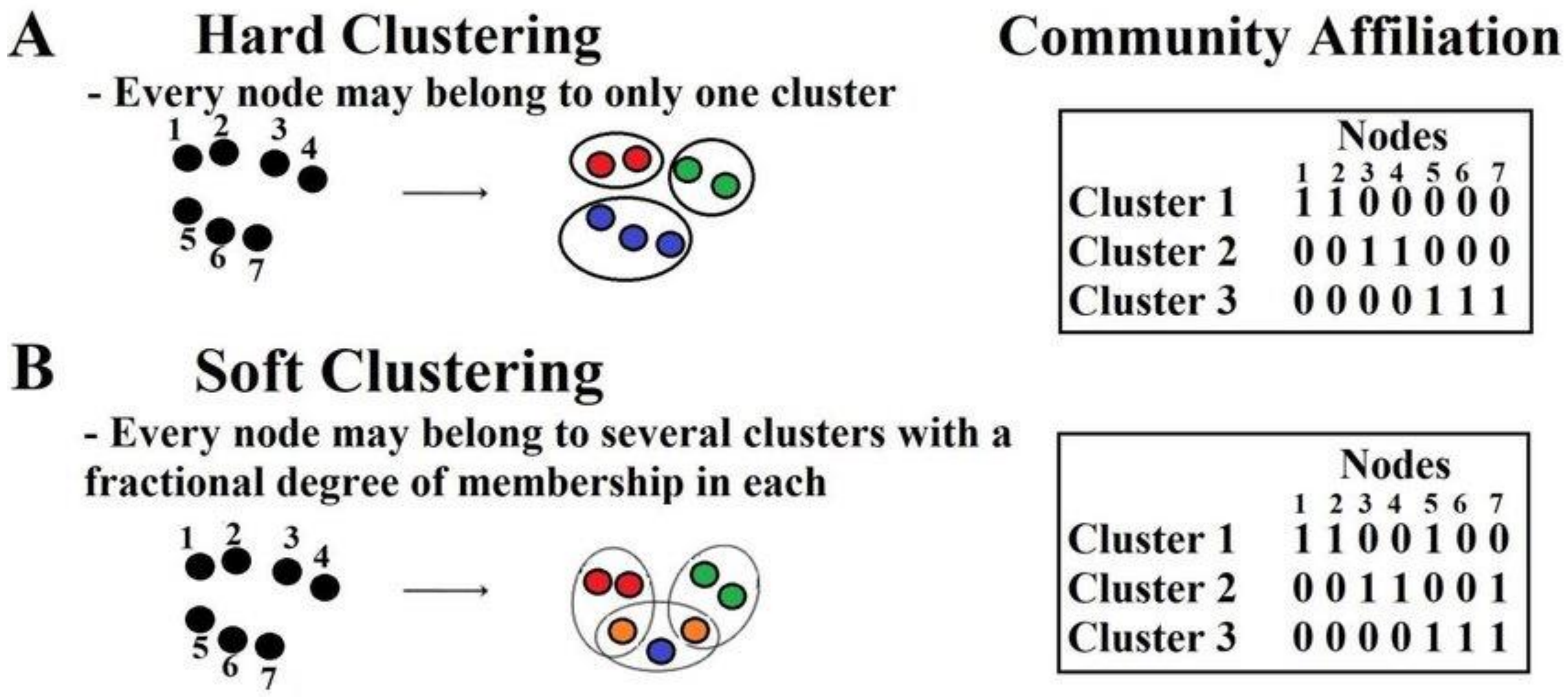
- 30% Coursework (group coursework)
- 70% exam (2 hours in-person invigilated exam)

COMP4139 Early Module Feedback

Hard Clustering vs soft clustering



Hard Clustering vs soft clustering



Mixture Models

- Probabilistic grounded way of doing soft clustering
- Each cluster would be a distribution model, generally a normal distribution
- This has unknown parameters: mean and variance
- How do we find these parameters?
- EM algorithm!

1-D Mixture Models

- Observations $x_1 \dots x_n$
  - K=2 Gaussians with unknown μ, σ^2
  - estimation trivial if we know the source of each observation

$$\mu_b = \frac{x_1 + x_2 + \ldots + x_{n_b}}{n_b}$$

$$\sigma_b^2 = \frac{(x_1 - \mu_1)^2 + \ldots + (x_n - \mu_n)^2}{n_b}$$

1-D Mixture Models

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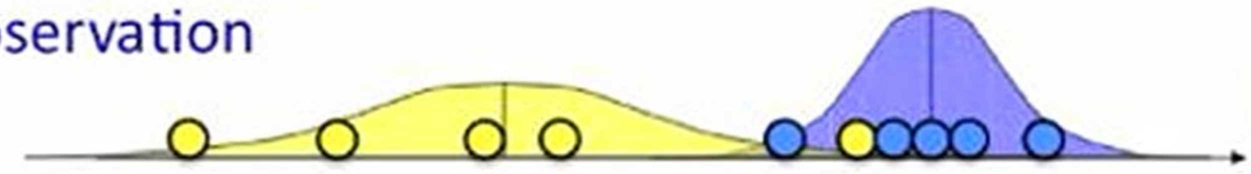
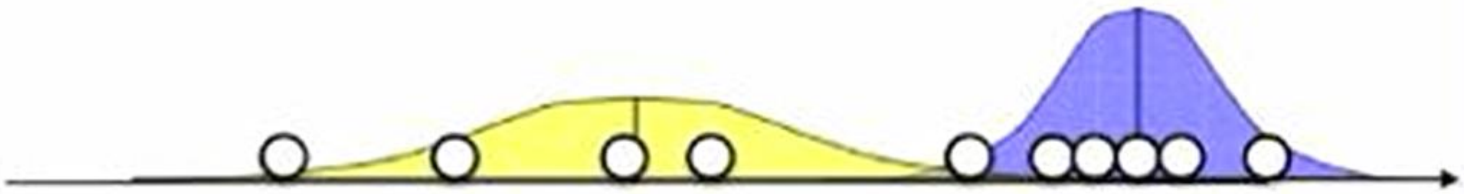


What if we do not know the source?

If we know parameters of the Gaussian, we can guess a or b.

$$P(b \mid x_i) = \frac{P(x_i \mid b)P(b)}{P(x_i \mid b)P(b) + P(x_i \mid a)P(a)}$$

$$P(x_i \mid b) = \frac{1}{\sqrt{2\pi\sigma_b^2}} \exp\left(-\frac{(x_i - \mu_b)^2}{2\sigma_b^2} \right)$$

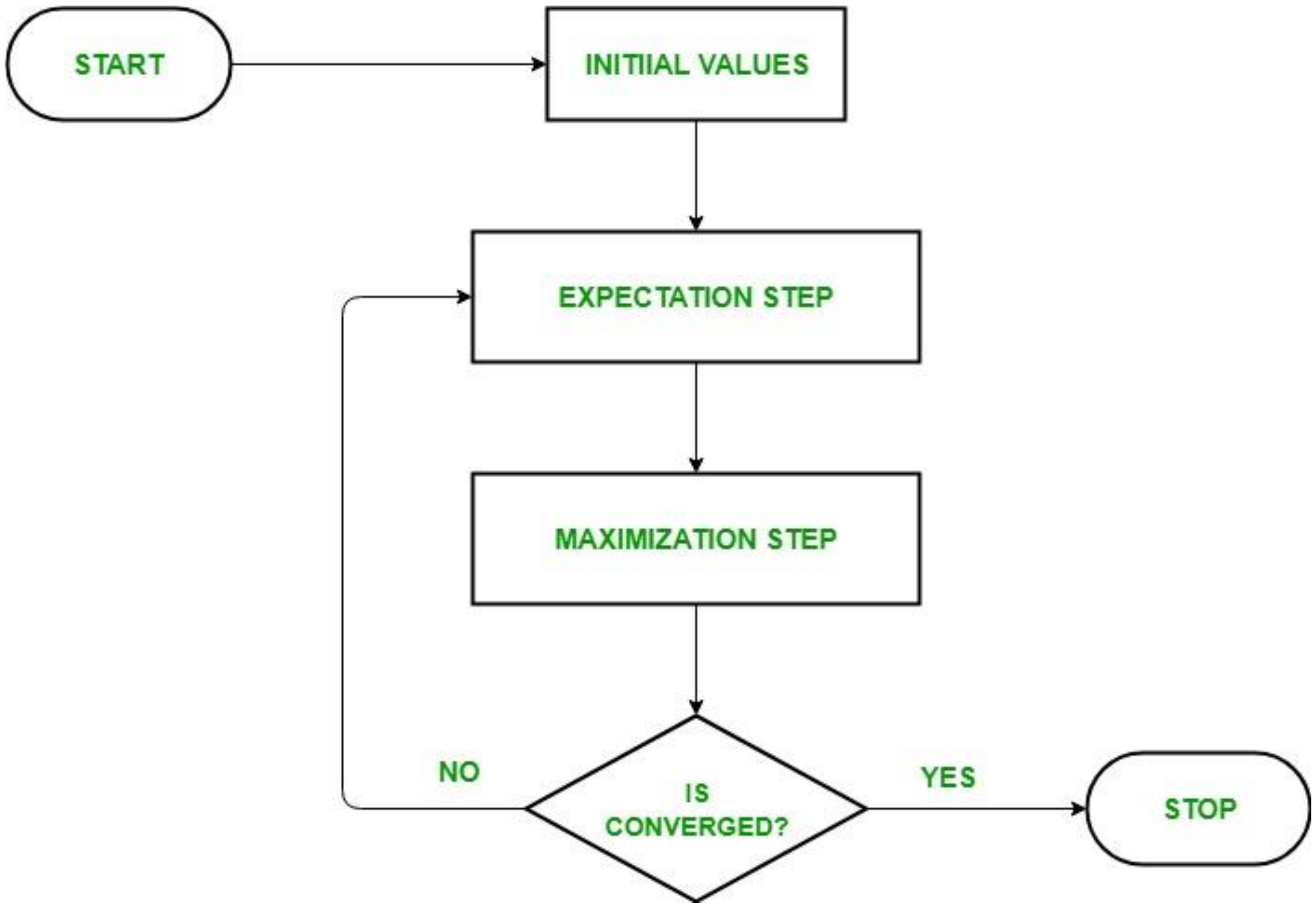


1-D Mixture Models

- Chicken and egg problem
  - need (μ_a, σ_a^2) and (μ_b, σ_b^2) to guess source of points
  - need to know source to estimate (μ_a, σ_a^2) and (μ_b, σ_b^2)

- EM algorithm
  - start with two randomly placed Gaussians (μ_a, σ_a^2), (μ_b, σ_b^2)
  - for each point: $P(b|x_i)$ = does it look like it came from b?
  - adjust (μ_a, σ_a^2) and (μ_b, σ_b^2) to fit points assigned to them

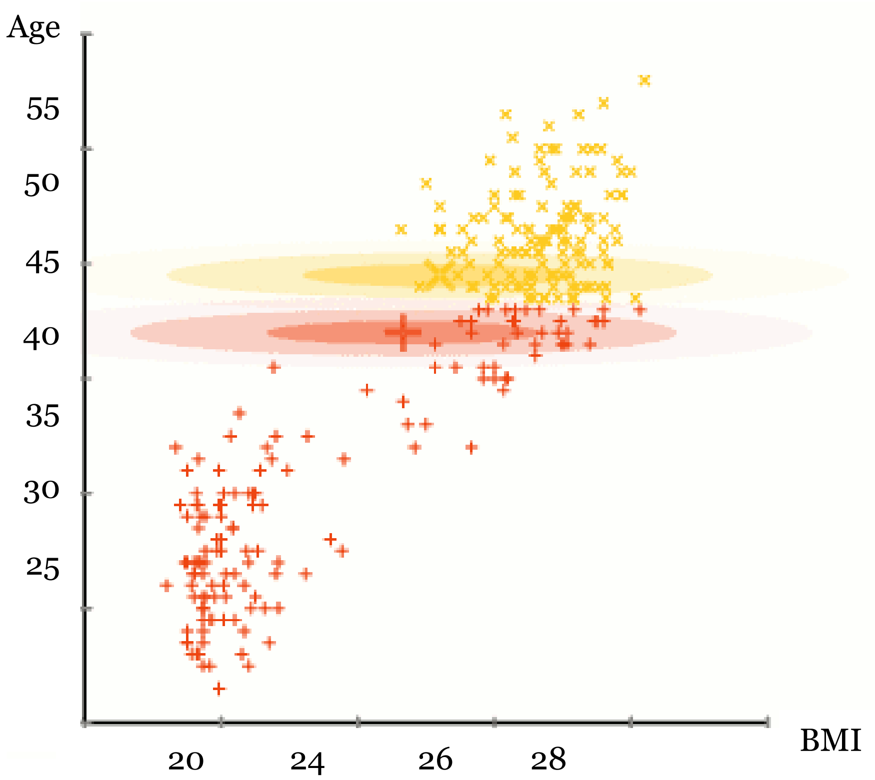
Expectation Maximisation



Expectation Maximisation

Algorithm:

- Assume a distribution model that describes the data (e.g. Gaussian)
- Initialise the model parameters (mean & variance)
- *Repeat*
  - E step: calculate the expected value (which class each data belongs to) of the log likelihood function with the current model parameters.
  - M step: estimate model parameters that maximise the log likelihood function.
- *Until* convergence.



Expectation Maximisation

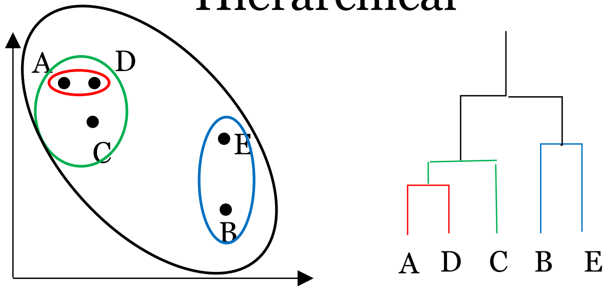
Pros/Cons

- ✓ Soft clustering
- ✓ Noise handling
- ✓ Effective for statistical models like GMMs
- × Restricted by the distribution model
- × Sensitive to initialisation
- × Need to specificity the number of clusters
- × Slow convergence

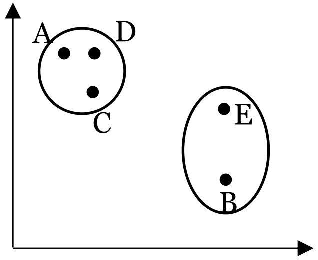
Categories of Cluster Analysis

- Hierarchical vs Partitional

Hierarchical

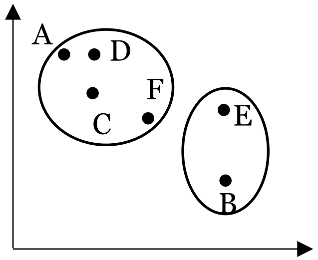


Partitional

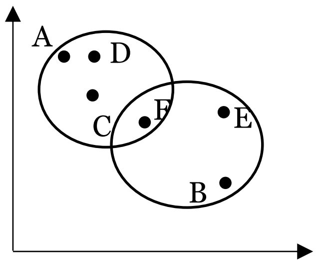


- Exclusive vs Overlapping vs Fuzzy

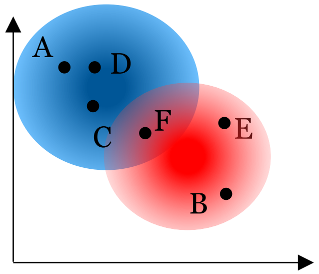
Exclusive



Overlapping

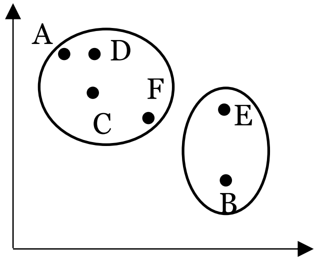


Fuzzy

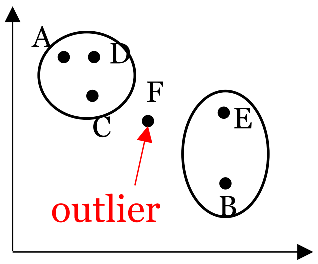


- Complete vs Partial

Complete



Partial



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Comparison of Different Methods

Algorithms	Category	Pros	Cons
Centroid model (K-means)	Partitional, Exclusive, Complete	• Simple and efficient	• Specify the number of clusters • Sensitive to outliers • Sensitive to initialisation
Connectivity model (Agglomerative Hierachical)	**Hierarchical**, Exclusive, Complete	• Flexible with number of clusters • Capture hierarchical relationship	• Requires larger memory and computationally expensive
Density model (DBSCAN)	Partitional, Exclusive, **Partial**	• Robust to outliers • Automatically determine number of clusters	• Not robust to variable density clusters • Computationally expensive • Sensitive to parameter settings
Distribution model (EM)	Partitional, **Fuzzy**, Complete	• Soft clustering	• Specify the number of clusters • Restricted by distribution model • Sensitive to initialisation

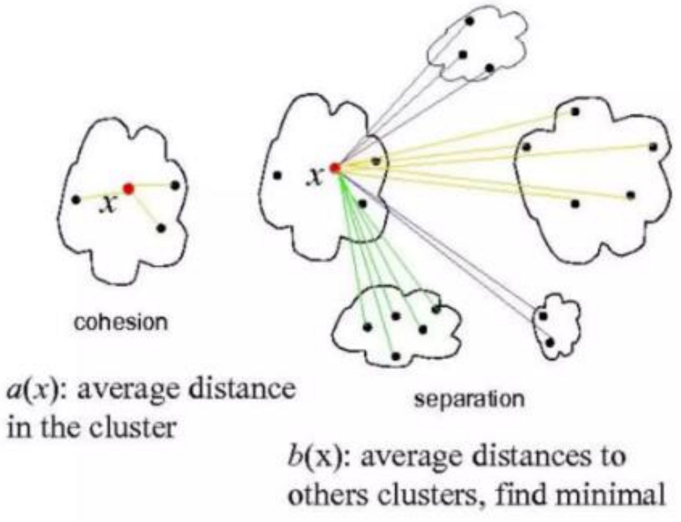
Evaluation of Cluster Analysis

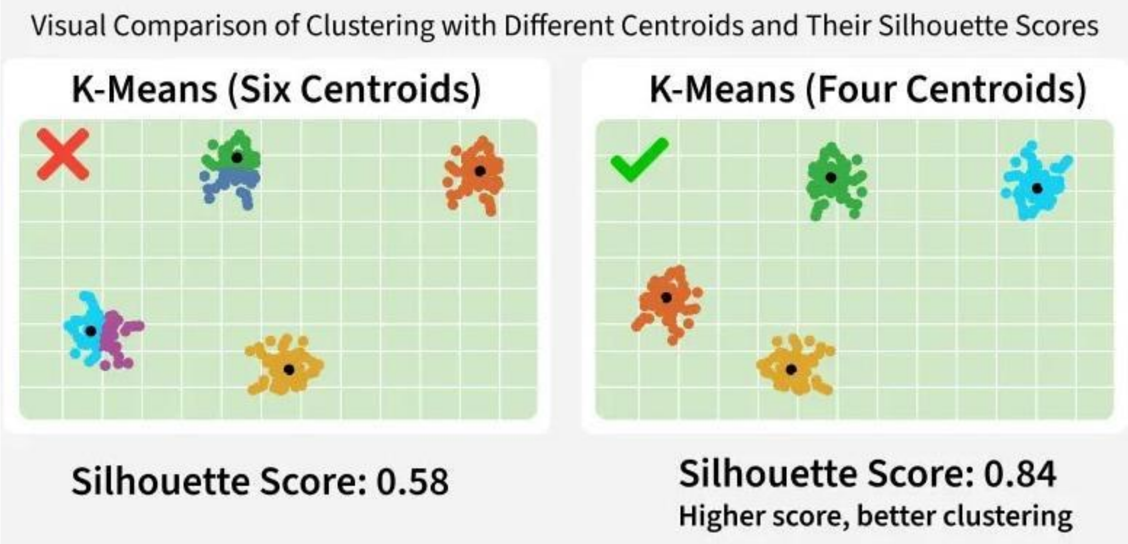
- If the clustering algorithm separates dissimilar data samples apart and similar data samples together, then it has performed well.

$$Sc = \frac{b-a}{\max(a,b)}$$

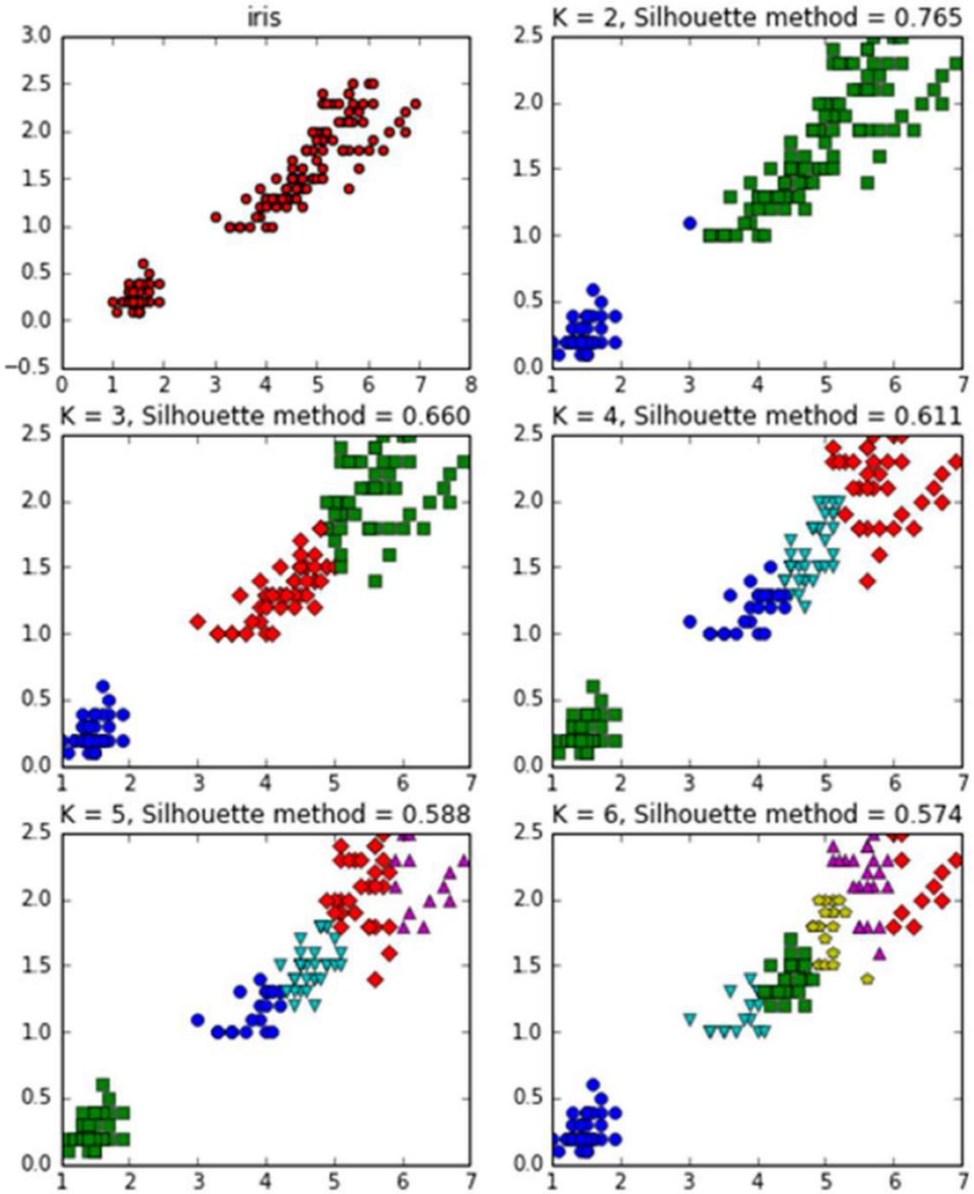
- Methods
  - Silhouette Coefficient (-1 to 1):

    -1 incorrect clustering, 0 overlapping clusters, 1 highly dense separated clusters.
    a: the mean distance between a sample and all other points in the same cluster
    b: the mean distance between a sample and all other points in the next nearest cluster





Another example



Evaluation of Cluster Analysis

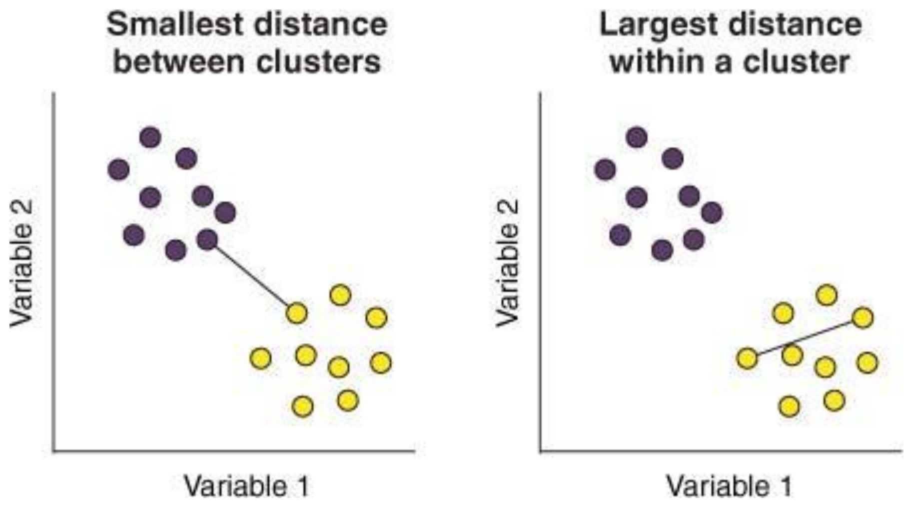
- If the clustering algorithm separates dissimilar data samples apart and similar data samples together, then it has performed well.

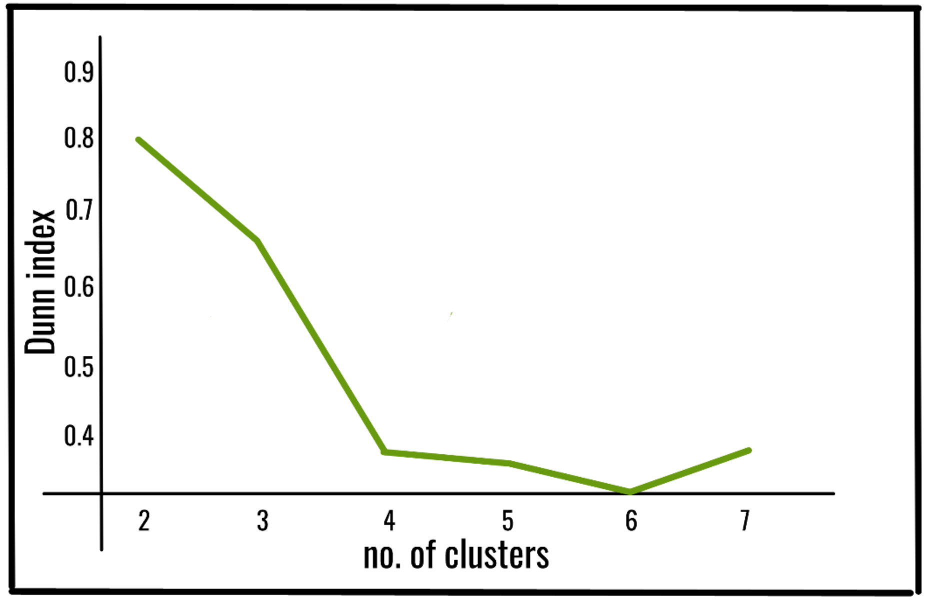
- Methods

  - Dunn's Index:

    d(i,j) represents the inter-cluster distance (centroids between two clusters) between i and j. d'(k) measures the intra-cluster distance of cluster k.

$$DI = \frac{\min_{1 \le i < j \le n} d(i,j)}{\max_{1 \le k \le n} d'(k)}$$





Difference between SC and DI

Feature	Silhouette Coefficient	Dunn Index
What it measures	For each data point, it measures how well it is clustered by comparing its average intra-cluster distance to its average inter-cluster distance.	The ratio of the minimum inter-cluster distance to the maximum intra-cluster distance.
Goal	A higher value indicates that the point is well-matched to its own cluster and poorly matched to neighboring clusters.	A higher value indicates more compact and better-separated clusters.
Range	$[-1, 1]$	$[0, \infty)$
Interpretation	Scores near $+1$ are excellent; scores near 0 indicate overlapping clusters; scores near -1 indicate the point may have been assigned to the wrong cluster.	A higher index value suggests better clustering quality.
Calculation	A per-point calculation that is then averaged across all points.	A single value calculated for the entire clustering solution, based on the minimum and maximum distances between all clusters.

Recap

- What is unsupervised learning?
- Similarity between vectors
- K-Means Clustering
- Agglomerative Hierarchical Clustering
- Density-based Spatial Clustering of Applications with Noise (DBSCAN)
- Expectation Maximisation
- Silhouette Co-efficient
- Dunn’s Index

What we will learn in the next lecture

- What is a Convolutional Neural Network?
- What is a convolution?
- What is max-pooling?
- How do we handle color images with neural networks?
- What is vanishing gradient?
- Assignment 2 discussion